# Rig day — 2026-10-22 (THURSDAY, ACUTE #4): verify the arrays first, then localize VPL, then the 10-14 plan

> **Built 2026-10-21 from `RIG_DAY_2026-10-14.md` + the 10-16 post-mortem (`LAB_NOTEBOOK_2026-10-16.md`, `docs/ACUTE4_LOCALIZATION_METHODS.md`).** Every command line below is the output of `& $py rig\rig_runcard.py --phase <phase> --day 2026-10-22 …` (lint-checked by `rig\lint_runbook.py`). Blanks `<resp> <c1>,<c2> <V> <blk>` are filled by passing `--resp 1,3,4,6 --ctrl 34,40 --baseline <V> --block <dir>` to the runcard, which prints the lines ready to paste and writes them to `day_2026-10-22\runcard_<phase>.txt`.
>
> **What acute #3 taught (10-14 → 10-16):** the 64-ch cortical array was ELECTRICALLY BROKEN before the first recording (banks ch1-16 and 49-64 dead, 21/64 contacts with their own signal, every channel = the reference) and nothing on the rig said so: the cortical gate graded a 63-channel common-mode touch wave as GO, and the thalamic depth ladder then read 0/8 at two positions on a readout that could not see. The thalamic target position is therefore still UNKNOWN (acute #2: 8/8 at the same coordinates, 7.8 mm).
>
> **What is new today (all built + known-answer-tested 10-21):** 1. `rig\array_health.py` — `ARRAY HEALTH: OK / SUSPECT / BROKEN` from ANY Wav1 recording, no trigger needed (own-signal contacts, pairwise corr, dead 16-ch banks). Acute-2 blocks → OK; every 10-14 block → BROKEN; the 10-13 in-air bench block → UNINFORMATIVE (mains). Runs on the SALINE bench block before induction and on the first in-tissue block. 2. `placement_gate --mode cortical` now prints the array line, the **common-mode touch wave** (median of all channels) and the **re-referenced focal map** (channel minus median) with a control-channel candidate. BROKEN array → **NO-GO**; common-mode wave > 20 µV AND > 50 % of the focal peak → **MARGINAL** with "move the reference". Acute-2 GO block still reads GO (+63 µV CM under a −219 µV focal response is fine); 10-14 block 204658 reads NO-GO (array broken; re-ref ch47 −35 µV z 50 is the real focal response). 3. `--newest` finds blocks nested as `Acute_102226\<blk>\<blk>` (10-14 needed `--block` on every line) and sorts by the `.tsq` time, not the copy time. 4. `rig_runcard --phase cgate` (cortical gate lines) and `--phase gate --short` (1.6-min `design_rungate_short.csv`, 18/25/30 µA, for the depth ladder; the 4.4-min full design once at the chosen depth). 5. `pull_blocks.ps1` defaults to `Desktop\Data\Acute_102226`.
>
> **NOT changed (deliberately, the night before):** no C++ rebuild. The controller still reads the RAW channel (`--feature-map`), no software re-reference. The reference must therefore be fixed PHYSICALLY (skull screw / quiet site) and proven by the cortical gate's common-mode line before the thalamic descent.

**Staffing:** plan as on 10-14 — mostly SOLO at the table; the engineer lane is `HELPER_CARD_2026-10-22.md`. **Hard stop on experiments 23:00.** Tier A → B → C is the plan; the cut list is for a late start; the early/late repeat (6g) is the LAST arm regardless and is never cut. **No depth-ladder clock stop today: the ladder runs to GO or to the end of the position list (G′), then the surgeon decides.**

**SOLO MODE (as 10-14): every block transfer is you breaking scrub, so transfer only what the NEXT decision needs and batch the rest.** - **Immediate (the check line reads the block):** the saline bench block (Phase 0), every cortical-gate block (G), every thalamic depth block (G′), the quiet block (Phase 1). - **Batched (the check line reads the capture/log on this PC):** thwack batteries, burst probe, gap, duty-ID, every arm, R sweep, drift — two batches: during the Phase-1 settle and at WRAP (burst ≈ 7 GB, start it first). The `ingest_block` lines for these phases run at WRAP in block order. - `run_audit`, `tracking_metrics`, `fit_burst_plant`, `check_dead_time`, `check_mimo_rank`, `analyze_r_sweep` never need the block: run them right after `Summary:` as written.

Conventions: 64 ch (`-InputChannels 64`), Microprobes pair table `--pairs "16:12,15:11,14:10,13:9,4:8,3:7,2:6,1:5"`, recording started LAST, stim enable ON at `go`, **nothing else runs on this PC during a run**, cpp-primary, uMax 30, blacklist from TODAY's Phase 1 only, y-liveness after every `go`, `run_audit` after every `Summary:`. Python = `C:\Users\brets\Documents\Repositories\PythonIntanAnalysis\.venv\Scripts\python.exe` (set `$py`; bare `python` is the Store stub). Every PowerShell: `Set-ExecutionPolicy -Scope Process Bypass` + `0_preflight` once. Blocks: `C:\Users\brets\Desktop\Data\Acute_102226\` (USB flash drive by hand, 2-3 min per block).

## Banked before the day

- Designs (`DESIGN OK`): `design_rungate_short.csv` 1.6 min (ladder) · `design_rungate.csv` 4.4 min (chosen depth) · `design_runburst1.csv` 15.9 min · `design_rungap1.csv` 2.2 min · `design_rundutyid_all8.csv` 5.0 min.
- Rig path as verified 10-12/10-13/10-14: USB dongle `Ethernet 2`, RZ2 10.1.0.100, `6_udp_selftest -LiveStim` 8/8, `cpp_controller --selftest` 17/17, bench 28/28. Delivery on 10-14 verified 720/720 + 240/240 (the WIRE is fine).
- Dead time = 3 ticks (09-10 prep) → fitter default `inputDelayTicks 2`; measure again today on the gate probe.

- 10-21: gate phase sim-rehearsed from the new runcard (`day_2026-10-22\rehearsal_gate_1021.log`).
- The 10-14 touch templates (`NNController outputs/.../touch/Acute_2026-10-14`) are RAW and common-mode dominated: **not targets for today**. Today's Phase-2 battery makes `Acute_2026-10-22`.

## 08:00 Phase 0 — bring-up AND array verification, BEFORE induction (50 min)

### 0a. Pre-induction checklist

| # | item | PASS | value / note |
|---|---|---|---|
| 1 | Stimulator battery charged; indicator checked at 08:00 | ☐ | |
| 2 | PZ2 amplifier ON and on its charger / battery plan for 14 h (09-10 PZ2-off cost 3 runs; 10-14 one VOID run) | ☐ | |
| 3 | Synapse stim safety button ON; stim-enable OFF until `go` | ☐ | |
| 4 | **Cortical array connector + headstage seated, strain-relieved; array inspected under magnification (10-16: 83 % of acute mechanical failures are connector-related)** | ☐ | |
| 5 | **Reference plan: cortical reference on a SKULL SCREW / quiet site (NOT on responsive tissue); thalamic array reference; animal ground. Written here:** cortical ref = _____ thalamic ref = _____ gnd = _____ | ☐ | |
| 6 | Electrode map: cortical grid = row-major, ch1-8 = row 1 (what `placement_gate` assumes); if a DIFFERENT array is used today write its map as a 64-entry JSON and pass `--map` (also `-InputChannels`) | ☐ | array today: _____ |
| 7 | **Impedance in SALINE, both arrays, ref + gnd wires submerged, no bubbles: Synapse impedance at 100 Hz AND 1 kHz → "Save Impedance to CSV" into the block folder.** Flag < 50 kΩ or > 2.5 MΩ, and any whole 16-ch bank at one value (Barrese 2013 / Caldwell 2018) | ☐ | outliers: _____ |
| 8 | **Saline bench block (60 s record, both arrays submerged) →** `array_health.py --newest --context saline` **= `ARRAY HEALTH: OK`.** BROKEN = reseat connector, re-record; still BROKEN = **swap the array before induction** (the day does not start on a broken readout) | ☐ | verdict _____ own-signal ___/64 corr _____ |
| 9 | **Thalamic array delivery into saline: run the wire gate (0d) with the array in saline; IZ2 front-panel LEDs GREEN on the stim bank during the run, never RED (red = ±10 V compliance = open lead)** | ☐ | pairs green ___/8 |
| 10 | Physiology monitoring per protocol: every 15 min HR + body temp on the PhysioSuite → `physio_timer -EveryMin 15` window open | ☐ | PhysioSuite on, probes placed: ☐ |
| 11 | This PC: no other apps, sleep off, ≥ 50 GB free (preflight prints both); never pipe terminal B through another program | ☐ | |
| 12 | USB flash drive empty, both PCs' USB ports known; drive letter here: _____ ; `Desktop\Data\Acute_102226\` exists | ☐ | |
| 13 | Synapse PC: sleep off, ≥ 50 GB free on the tank drive, project = the frozen 64-ch circuit with Scle/Plse/sSig/UDP1/Wav1/Wav2 stores, tank `Acute_102226` | ☐ | |
| 14 | **Thalamic coordinates + REFERENCE PLANE written down** (acute #2 GO: AP _____ ML _____ depth 7.8 mm from _____ [skull / dura / brain surface], tip-length ladder 7.4-8.0 mm; literature forepaw VPL 5-7 mm from the brain surface at AP −2.4…−2.7, ML 2.5-2.8) | ☐ | |
| 15 | Optional per protocol / surgeon's call: DiI on the thalamic shank backs before insertion; end-of-day marking lesion | ☐ | |

### 0b. Network + loop (`rig_runcard --phase phase0`)

```
Set-ExecutionPolicy -Scope Process Bypass; $py = 'C:\Users\brets\Documents\Repositories\PythonIntanAnalysis\.venv\Scripts\python.exe'
Get-PnpDevice | Where-Object { $_.InstanceId -match 'VEN_4550' } | Format-List FriendlyName, Status, Problem   # PO8e must be OK
.\rig\0_preflight.ps1 -InputChannels 64            # PASS: droppedControlTicks=0; also prints sleep setting + free space
.\rig\net_diag.ps1                                  # auto-detects the dongle ('Ethernet 2'); PASS: the path is correctly configured
& $py rig\find_rz2.py                              # exactly one device: 10.1.0.100
.\rig\6_udp_selftest.ps1 -RZ2 10.1.0.100 -LiveStim  # 8/8 PASS; E2 live packet is a ZERO -- before the animal is on the table
```

### 0c. Block transfer (MANUAL, USB flash drive) + monitoring window

Copy the new block folder from the Synapse tank onto the drive, carry it over, copy it into `C:\Users\brets\Desktop\Data\Acute_102226\` (blocks are `<subject>-261022-<hhmmss>`; nested one level deeper is fine now). Every gate/ingest line uses `--newest` = the newest `.tsq` under `Desktop\Data`; **check the `newest block:` line is the recording you just stopped.**

```
Start-Process powershell -ArgumentList '-NoExit','-Command',".\rig\physio_timer.ps1 -Day 2026-10-22 -EveryMin 1
5"   # HR + body temp every 15 min
.\rig\pull_blocks.ps1 -Source <drive>:\Acute_102226 -Once      # instead of Explorer: mirrors only NEW blocks, p
rints NEW BLOCK: <dir>
```

### 0d. Card probe + SALINE array verification (the new gate of the day)

```
.\MpcPo8eUdpClosedLoop.exe 10.1.0.100 . 64 --controller constant --constant-output 0 --skip-udp-send --max-contr
ol-ticks 300   # Streaming. numChannels=64, 300/300, window 6
# both arrays in saline (ref + gnd submerged): impedance 100 Hz + 1 kHz -> CSV (checklist 7); then a 60 s block
-> drive -> here:
& $py rig\array_health.py --newest --context saline --n-channels 64 --label "cortical saline bench"      # ARRAY
HEALTH: OK
# thalamic delivery into saline = wire gate on the short design (1.6 min), watch the IZ2 LEDs (green, never re
d):
& $py rig\rig_runcard.py --phase gate --short --depth 0 --day 2026-10-22     # prints A / B / check lines with -
Run gate0
# after Summary: run_audit -> AUDIT gate0: PASS; check_impulse_delivery ... --design-meta design_rungate_short_m
eta.json -> DELIVERY VERIFIED, 0 missed
```

`ARRAY HEALTH: OK` + `DELIVERY VERIFIED` + green LEDs = both arrays are proven BEFORE the animal. BROKEN → checklist 8.

## 09:30 Induction → surgery (solo lane). Target first recording 15:30; solo-realistic 16:30.

Both craniotomies in ONE drilling sitting. **Skull screw for the cortical reference placed FIRST and wired (checklist 5).** Cortical array aimed at the acute-#2 focus (grid rows 5-6, cols 2-7; best channels 34/39/40/47). Thalamic target = acute-#2 coordinates and plane (checklist 14); the depth ladder below starts there.

## Phase G — cortical placement gate (repeat until GO, ~5-6 min per cycle incl. the drive round-trip)

Seat the planar array → **40 thwacks on D1** (~45 s) → stop recording → round-trip (0c) → `& $py rig\rig_runcard.py --phase cgate --day 2026-10-22` prints:

```
& $py rig\array_health.py --newest --context tissue --n-channels 64 --label "cortical in tissue"   # ARRAY HEALT
H: OK first
& $py rig\placement_gate.py --mode cortical --newest   # CORTICAL GATE: GO; also read the 'array', 'common-mode'
and 're-referenced' lines
& $py rig\ingest_block.py --label "D1 gate thwack" --mode thwack --day 2026-10-22
```

**GO** = z ≥ 8, split-half ≥ 0.80, latency 8-45 ms, AND array OK, raw footprint ≤ 32 ch, common-mode wave not flagged. Read three lines aloud every cycle: `array: …` · `common-mode touch wave …` · `re-referenced best ch….` - `ARRAY BROKEN` → **NO-GO regardless of z**: reseat the connector/headstage and re-thwack; if unchanged, swap the array. The thalamic ladder cannot be read on a broken array (that was 10-14). - common-mode flagged (> 20 µV and > 50 % of the focal peak) → **the reference sits on responsive tissue: move it (skull screw), re-thwack 40×**. Do not start the thalamic ladder with a flagged reference: the control channel would be the reference site. - MARGINAL/NO-GO on z → read the ADJUST line (edge → shift toward it; no focus → move/re-aim), re-thwack, re-run. Log: cycle ___ verdict _____ z _____ array _____ CM _____ µV re-ref best ch _____ (→ control-channel candidate `<c1>`). Never the speaker.

## Phase G′ — thalamic placement gate (depth ladder on the SHORT design, ~4 min per position incl. transfer)

**Order: `CORTICAL GATE: GO` (array OK, reference clean) BEFORE grading any thalamic position** — the thalamic verdict is read off the cortical array. Re-thwack 40× after any cortical move. Position list (write the plane from checklist 14; depths are to the TIP of the 7.4-8.0 mm ladder): | d | position | why | |---|---|---| | 1 | acute-#2 coordinates, 7.8 mm | the only position that ever gave 8/8 (09-10) | | 2 | same AP/ML, 7.0 mm | literature forepaw VPL 6-7 mm (Choi 2016 6-7 mm; Aguilar 2008 5-7 mm) | | 3 | same AP/ML, 6.5 mm | | | 4 | same AP/ML, 6.0 mm | | |

5+ | move AP ±0.3 / ML ±0.3 at the best depth so far; (if an audio monitor is available: forepaw tap while advancing from 5.5 mm, units at 6-7 mm, tactile latency 5-9 ms) | | `& $py rig\rig_runcard.py --phase gate --short --day 2026-10-22 --depth <d>` prints (d = 1, 2, 3, …):

```
# A:
.\cpp_controller.exe --mode openloop --play design_rungate_short.csv --output-count 8 --capture capture_rig_rungate<d>.csv
# B:
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run gate<d> -Design design_rungate_short.csv -Capture capture_rig_rungate<d>.csv
# after Summary: (recording stopped, round-trip done)
& $py rig\run_audit.py --run gate<d> --loop-log loop_rungate<d>.log --capture capture_rig_rungate<d>.csv --design design_rungate_short.csv
& $py rig\placement_gate.py --mode thalamic --newest --min-amp 13
& $py rig\check_impulse_delivery.py --block <blk> --capture capture_rig_rungate<d>.csv --amps 18 25 30 --pairs "16:12,15:11,14:10,13:9,4:8,3:7,2:6,1:5" --design-meta design_rungate_short_meta.json   # wire gate re-bank
& $py rig\check_dead_time.py --capture capture_rig_rungate<d>.csv --y-channel <best ch> --json day_2026-10-22\dead_time_gate<d>.json   # -> inputDelayTicks for the fit
& $py rig\ingest_block.py --label "gate depth <d> probe" --day 2026-10-22
```

During every probe run: **IZ2 LEDs green on the stim bank, never red** (red = compliance = a lead has left tissue / is open; the command monitor `DELIVERY VERIFIED` cannot see that — Wav1 shows no stim artifact either way). GO = ≥ 2 RESPONSIVE pairs AND footprint eff rank ≥ 2. Tie-break: more responsive pairs, then higher eff rank, then lower median footprint corr. **At the chosen position run the FULL design once** (`rig_runcard --phase gate --day 2026-10-22 --depth <d+1>` without `--short`, 4.4 min): it banks the knees, dead time and the wire gate at 13/18/25 µA for the fit. Depth _____ responsive pairs _____ eff rank _____ best pair _____ best ch _____ dead-time onset _____ (→ `--delay-ticks` = onset−1). **If the whole list is NO-GO with a healthy array and clean reference:** that is a real placement result this time. Surgeon's call: continue AP/ML search, or bank the thwack battery (Tier A Phase 2, no stim needed) and stop.

## Phase 1 — seal, settle 30 min; quiet capture (60 s) + blacklist + baseline

```
& $py rig\ingest_block.py --label quiet --day 2026-10-22      # runs analyze_quiet_capture -> galleries/<blk>/quiet.json
```

blacklist _____ baseline _____ V (→ `--blacklist` and `--baseline` on every later runcard). Eat/drink here. Batch 1 of the block transfers (gate + quiet blocks are already over).

# ========== TIER A (minimum publishable day, ~3.5 h) ==========

## Phase 2 — thwack battery: 100/site × 6 sites + SHAM (~40 min)

Sites by achievability from the gate focus (acute #2: P1 MP LP P3 strongest; add D1 + one digit). Per block (`rig_runcard --phase thwack --site <SITE> --block <blk>`):

```
& $py rig\ingest_block.py --label "<SITE> thwack" --mode thwack --day 2026-10-22      # SOLO: batched (block at WRAP)
& $py C:\Users\brets\Documents\Repositories\NNController\scripts\touch\extract_nthw_templates.py --block <blk> --site <SITE> --day 2026-10-22 --min-pulse-ms 50 --n-channels 64
```

If first recording slipped past 17:30: 4 sites + SHAM. Templates are built from the control channel `<c1>`; if the common-mode line was non-zero but not flagged, the gate JSON's `reref` block has the re-referenced map for the offline rebuild.

## Phase 3 — burst-ladder probe (16 min) + gap scan (2 min)

`rig_runcard --phase burst --day 2026-10-22 --blacklist <list> --baseline <V>`:

```
.\cpp_controller.exe --mode openloop --play design_runburst1.csv --output-count 8 --capture capture_rig_runburst
1.csv
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run burst1 -Design design_runbur
st1.csv -Capture capture_rig_runburst1.csv -Baseline <V>
& $py rig\run_audit.py --run burst1 --loop-log loop_runburst1.log --capture capture_rig_runburst1.csv --design d
esign_runburst1.csv
& $py rig\check_impulse_delivery.py --block <blk> --capture capture_rig_runburst1.csv --amps 6 9 13 18 25 --pair
s "16:12,15:11,14:10,13:9,4:8,3:7,2:6,1:5" --design-meta design_runburst1_meta.json
& $py rig\fit_burst_plant.py --capture capture_rig_runburst1.csv --meta design_runburst1_meta.json --json day_20
26-10-22\burst_plant.json --png day_2026-10-22\burst_plant.png
& $py rig\placement_gate.py --mode thalamic --newest --min-amp 13 --blacklist <list>      # knees per pair
& $py rig\ingest_block.py --label "burst ladder probe" --day 2026-10-22      # SOLO: batched (block ~7 GB, at WR
AP)
```

Gap scan on the gate's BEST pair (`rig_runcard --phase gap --day 2026-10-22 --gap-pair <best pair> --gap-amp 25` regenerates and validates `design_rungap1.csv`; without `--gap-pair` the pre-built pair-4 design plays):

```
.\cpp_controller.exe --mode openloop --play design_rungap1.csv --output-count 8 --capture capture_rig_rungap1.cs
v
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run gap1 -Design design_rungap1.
csv -Capture capture_rig_rungap1.csv -Baseline <V>
& $py rig\run_audit.py --run gap1 --loop-log loop_rungap1.log --capture capture_rig_rungap1.csv --design design_
rungap1.csv
& $py rig\fit_burst_plant.py --capture capture_rig_rungap1.csv --meta design_rungap1_meta.json --json day_2026-1
0-22\gap_scan.json --png day_2026-10-22\gap_scan.png
```

**operating gap** = `OPERATING GAP (first ratio >= 0.80)`: _____ ticks (default 24 if unreadable).

## Phase 4 — duty-cycled operating-point ID (5 min) → MIMO fit → rank gate

`rig_runcard --phase dutyid --day 2026-10-22 --resp <resp> --gap-ticks <operating gap> --baseline <V>` (regenerates `design_rundutyid.csv` + meta on the responsive pairs with the MEASURED gap; must print `DESIGN OK`), then:

```
.\cpp_controller.exe --mode openloop --play design_rundutyid.csv --output-count 8 --capture capture_rig_rundutyi
d.csv
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run dutyid -Design design_rundut
yid.csv -Capture capture_rig_rundutyid.csv -Baseline <V>
& $py rig\run_audit.py --run dutyid --loop-log loop_rundutyid.log --capture capture_rig_rundutyid.csv --design d
esign_rundutyid.csv --active-u u1,u3,u4,u6
& $py rig\ingest_block.py --label "duty-ID probe" --day 2026-10-22
# fit: rig_runcard --phase fit --day 2026-10-22 --resp <resp> --ctrl <c1>,<c2> --delay-ticks <onset-1>
matlab -batch "cd('C:\Users\brets\Documents\Repositories\TDTClosedLoop'); r = fit_sysid_from_capture('capture_ri
g_rundutyid.csv', struct('useInputs',[<resp>],'useOutputs',[<c1> <c2>],'orders',1:3,'inputDelayTicks',<onset-1
>)); export_plant_lti('plant_duty.lti', r.sys)"
& $py rig\check_dead_time.py --capture capture_rig_rundutyid.csv --u-channels <resp> --y-channel <c1> --lti plan
t_duty.lti   # VERDICT must be OK/centred
& $py rig\check_mimo_rank.py --model plant_duty.lti
.\cpp_controller.exe --mode mpc --model plant_duty.lti --target 0 --check-model   # Observer: ACTIVE, pole < 1,
DC gain printed; exits by itself
```

GO: per-output |corr| > 0.1 on ≥ 2 channels AND rank gate PASS → control channels _____ (channels on DIFFERENT pair footprints — the gate PNG shows them; **prefer channels the cortical gate's re-referenced map ranked, never a channel without its own signal**). FAIL → p = 1 on the best channel, pairs = top-2 by knee; the day still yields burst-vs-tonic (6c). `export_plant_lti(path, r.sys)` leaves `AllModels.mat` alone.

## Phase 5 — references + tapes + manifest (≤ 10 min)

`rig_runcard --phase refs --day 2026-10-22 --resp <resp> --ctrl <c1>,<c2> --baseline <V>`:

```
& $py rig\build_interleaved_run.py --templates-dir C:\Users\brets\Documents\Repositories\NNController\outputs\Bi
omimeticInversion\touch\Acute_2026-10-22 --sites LP P1 MP P3 D1 SHAM --channels <c1> <c2> --baseline <V> --model
plant_duty.lti --runs 3 --events 100 --total-ticks 22200 --umax 30 --pairs <resp> --out-dir day_2026-10-22
& $py training\closed_loop_sim.py --plant plant_duty.lti --ticks 2000 --pairs <resp> --feature-map <c1>,<c2> --o
utput-count 8 --launch --launch-args --pairs <resp> --feature-map <c1>,<c2> --q-weight 1e12,1e12 --r-weight 100,
100 --control-horizon 20 --umax 30 --reference day_2026-10-22\ref_mix_r1.csv --capture day_2026-10-22\capture_si
m_refcheck.csv
```

→ `wrote 3 run(s)`; sim: `Command varies` and the per-output means move toward the reference.
(`build_interleaved_run` REFUSES to overwrite an existing `ref_mix_r<N>.csv`: later refs use `--run-offset`.)

## Phase 6a — r1 MPC → r1 Choi (paired schedule; ~10 min incl. scoring)

`rig_runcard --phase arm --arm mpc --run 1 --day 2026-10-22 --resp <resp> --ctrl <c1>,<c2> --baseline <V>` (then `--arm choi`):

```
# MPC, terminal A; then B (recording LAST, enable ON at go; the y-liveness window opens and BEEPS on DEAD):
.\cpp_controller.exe --mode mpc --model plant_duty.lti --output-count 8 --pairs <resp> --feature-map <c1>,<c2> --q-weight 1e12,1e12 --r-weight 100,100 --control-horizon 20 --umin 0 --umax 30 --reference day_2026-10-22\ref_mix_r1.csv --capture day_2026-10-22\capture_mpc_mixr1.csv --max-packets 22200
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run mpc_mixr1 -Ticks 22200 -Capture day_2026-10-22\capture_mpc_mixr1.csv -Baseline <V>
# Choi:
.\cpp_controller.exe --mode openloop --play day_2026-10-22\design_choi_mix_r1.csv --output-count 8 --capture day_2026-10-22\capture_choi_mixr1.csv
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run choi_mixr1 -Ticks 22200 -Capture day_2026-10-22\capture_choi_mixr1.csv -Baseline <V>
# after each Summary:
& $py rig\run_audit.py --run <arm>_mixr1 --loop-log loop_run<arm>_mixr1.log --capture day_2026-10-22\capture_<arm>_mixr1.csv --ref day_2026-10-22\ref_mix_r1.csv --active-u u1,u3,u4,u6 --baseline <V>
& $py rig\tracking_metrics.py --capture day_2026-10-22\capture_<arm>_mixr1.csv --ref day_2026-10-22\ref_mix_r1.csv --y-channels <c1> <c2> --label <arm>_r1 --json day_2026-10-22\tracking_<arm>_r1.json --png day_2026-10-22\tracking_<arm>_r1.png
& $py rig\ingest_block.py --label "<arm> r1 arm" --mode arm --schedule day_2026-10-22\schedule_mix_r1.json --day 2026-10-22      # SOLO: batched (block at WRAP)
```

`AUDIT <arm>_mixr1: PASS` = fresh@≤100, dropped 0, observer ACTIVE pole<1, offsets=applied, y LIVE, u moves. An MPC arm with `--dist-observer` is a separate, labelled arm (offset-free MPC; not Choi-comparable). A small-ρ arm (`--effort-ratio ≤ 0.01`) adds `--qp-iters 5000` (10-13 sweep); R=100 arms keep the default.

**DECISION POINT A (clock ~19:30, or whenever Tier A is banked): tired → jump to WRAP. Else Tier B.**

# ========== TIER B (+1.5 h): 6b → 6c → 6d ==========

## 6b — decoupled-target MIMO run (only if the rank gate PASSED)

`& $py rig\build_decoupled_ref.py --ref day_2026-10-22\ref_mix_r1.csv --out day_2026-10-22\ref_decoupled_r1.csv` → MPC as 6a with `--reference …ref_decoupled_r1.csv --capture … capture_mpc_decoupled.csv` (`-Run mpc_decoupled`); score per the sidecar `_schedule.json`.

## 6c — burst-vs-tonic paired arms (schedule r1, two MPC runs back to back)

Run A = 6a's MPC (`--umin 0`); run B = same model/refs with `--umin <knee uA>`: `rig_runcard --phase arm --arm mpc --run 1 --suffix _umin --umin <knee>`. Report r0, charge ABOVE HOLD, and the Phase-7 drift probe for both.

## 6d — r2 MPC → r2 Choi (`--run 2`).

**DECISION POINT B (~21:00): tired → WRAP. Else Tier C.**

# ========== TIER C (+1 h): 6e → 6f ==========

## 6e — interleaved R sweep: ONE 13-min run, R ∈ {100, 30, 10} in 20-event blocks

```
& $py rig\build_interleaved_run.py --templates-dir C:\Users\brets\Documents\Repositories\NNController\outputs\BiomimeticInversion\touch\Acute_2026-10-22 --sites LP P1 MP P3 D1 SHAM --channels <c1> <c2> --baseline <V> --model plant_duty.lti --runs 1 --events 120 --total-ticks 26600 --umax 30 --pairs <resp> --out-dir day_2026-10-22 --run-offset 3
& $py rig\rig_runcard.py --phase rsweep --run 4 --day 2026-10-22 --resp <resp> --ctrl <c1>,<c2> --baseline <V>
# prints build_r_schedule, A (with --r-schedule), B, run_audit, analyze_r_sweep
```

`R SCHEDULE OK`; after the run every row of the `analyze_r_sweep` table says `rW0 ok`.

## 6f — r3 MPC → r3 Choi (`--run 3`). NN clone arm not prepared; skip.

# ========== WRAP (starts no later than 22:15, every tier) ==========

## 6g — EARLY/LATE repeat of r1, both arms, identical tapes/refs (~8 min). Never cut.

`rig_runcard --phase arm --arm mpc --run 1 --suffix _late …` and `--arm choi … --suffix _late`.

## Phase 7 — drift re-probe (duty-ID replay, 2 min) → stim zero → archive

`rig_runcard --phase drift …`:

```
.\cpp_controller.exe --mode openloop --play design_rundutyid.csv --output-count 8 --capture capture_rig_rundutyi
d_late.csv --max-packets 12000
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run dutyid_late -Ticks 12000 -Ca
pture capture_rig_rundutyid_late.csv -Baseline <V>
& $py rig\run_audit.py --run dutyid_late --loop-log loop_rundutyid_late.log --capture capture_rig_rundutyid_lat
e.csv --expect-ticks 12000
& $py rig\check_dead_time.py --capture capture_rig_rundutyid_late.csv --u-channels <resp> --y-channel <c1> --lti
plant_duty.lti   # gain/lag drift vs the fit: ______
.\rig\stim_zero.ps1                                  # 8 words x 2 s of zeros
.\rig\archive_day.ps1 -Day 2026-10-22                # zip of captures/logs/ledger/galleries -> Desktop\Data\Ar
chive
```

Batch 2 of the block transfers (burst first); the batched `ingest_block` lines run here in block order. **End-of-day array check (30 s):** `& $py rig\array_health.py --newest --context tissue --label "end of day"` on the last block → tells tomorrow's post-mortem whether the array survived the day. **23:00 hard stop.** Ledger rows are auto-appended by `ingest_block` into `BLOCK_LEDGER_2026-10-22.md`. Tomorrow: copy the archive zip + `Desktop\Data\Acute_102226` to the external drive; commit + push both repos (10-14, 10-16 and 10-21 work are all still uncommitted).

## Cut list (one list; cut from the bottom)

Tier A (6a + 6g) is never cut. If first recording > 17:30: battery 4 sites + SHAM, Tier A only + 6g. If > 18:30: skip the gap scan (use gap 24) and run 6a + 6g only. Otherwise cut in this order: 6f (r3) → 6e (R sweep) → 6d (r2) → 6c (burst-vs-tonic) → 6b (decoupled). **If the thalamic ladder ends NO-GO on a healthy array:** Phase 2 battery (all sites + SHAM, re-referenced templates are then the deliverable) → stim zero → archive. Do not run stim arms without a GO.

## Red flags (say them immediately)

- `ARRAY HEALTH: BROKEN` or the gate's `array:` line says BROKEN → stop; connector/headstage; swap the array.
- The gate's `common-mode touch wave` line is flagged → the reference is on responsive tissue; move it before G′.
- IZ2 LED RED during a probe run → compliance: a lead is open / out of tissue. Stop, inspect.
- The liveness window BEEPS / says `DEAD` (or `run_audit` says `y=DEAD`) → amplifier/PZ2 off. Stop the run.
- `AUDIT …: FAIL` → the next `2_loop` refuses to start until you type `override`; the surgeon decides.
- **Any crash or abort of a stim run** → **`.\rig\stim_zero.ps1` first**, then diagnose.
- B never prints `policy=fresh` by tick ~100 → Ctrl+C in B, restart B, `go` again.
- `Skipping … to position` in B → Synapse recording was started too early; stop, restart B, start recording LAST.
- A gate/ingest line prints `newest block:` with a time that is not the recording just stopped → the copy has not landed; it graded a STALE block.
- Never re-run Synapse Detect mid-day (08-29); never start another program on this PC during a run.

## Endurance rules

Water/food during the 16-min burst probe and the 30-min settle; sit for all arm runs; a helper runs the HELPER_CARD lane whenever present; the physio timer window is the clock for the animal.